AI INTELLIGENCE PLATFORM

INDIAGROUND

News Bias and Fact-Checking Platform for India

A dual-pipeline intelligence system for the Indian English news ecosystem

The Problem

Why Indian News Consumers Need Better Tools

Indian news consumers face a serious information gap. Existing verification systems were not built for the speed, complexity, or local context of the Indian English news ecosystem.

Who is affected

Indian readers lack localized credibility assessments.

- Western tools like Ground News focus primarily on U.S. and European outlets.
- Indian readers are left without localized credibility assessments.

Where current media verification falls short

Static systems cannot keep up with unfolding news.

- Existing systems depend on static ratings.
- They cannot detect bias in real time.
- They are unable to deliver dynamic, context-aware analysis as news develops.

The Methodological Innovation

IndiaGround replaces static credibility checks with a dual-pipeline intelligence system designed for live, context-aware news verification.

TRADITIONAL APPROACH

Western Approach

Focuses primarily on sentiment and tone

- Static credibility ratings
- United States and European focus
- Single-dimensional analysis

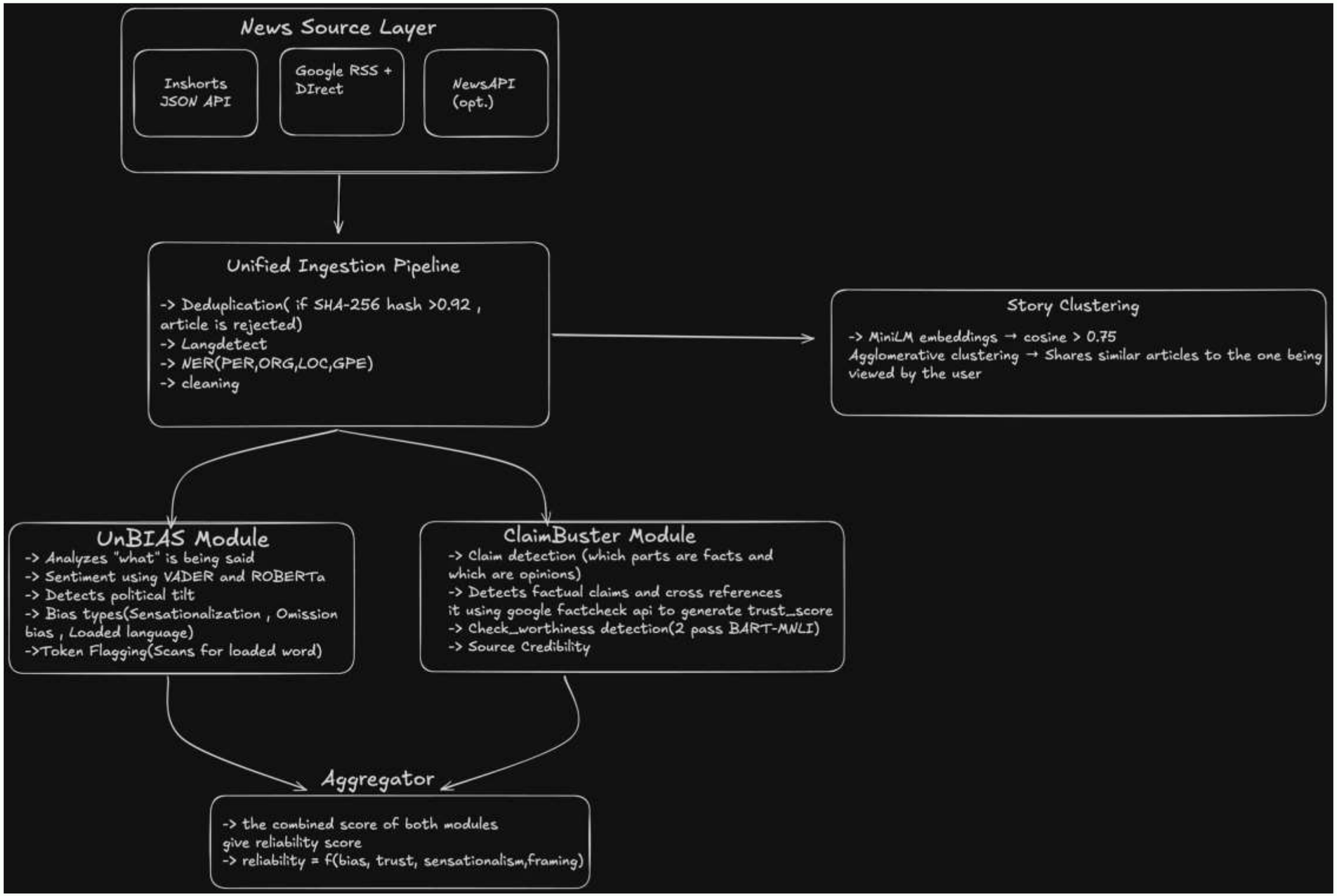
INDIAGROUND

IndiaGround Approach

Uses a dual-pipeline intelligence system

- Real-time bias detection
- India-specific media dictionary
- Multi-dimensional verification

Most current systems rely solely on sentiment analysis, but IndiaGround uses a dual-pipeline architecture that evaluates both HOW news is written and WHAT it claims.


News Source Layer
Inshorts JSON API
Google RSS + DIrect
NewsAPI (opt.)
Unified Ingestion Pipeline
-> Deduplication(if SHA-256 hash >0.92 , article is rejected)
-> Langdetect
-> NER(PER,ORG,LOC,GPE)
-> cleaning
Story Clustering
-> MiniLM embeddings → cosine > 0.75
Agglomerative clustering → Shares similar articles to the one being viewed by the user
UnBIAS Module
-> Analyzes "what" is being said
-> Sentiment using VADER and ROBERTa
-> Detects political tilt
-> Bias types(Sensationalization , Omission bias , Loaded language)
->Token Flagging(Scans for loaded word)
ClaimBuster Module
-> Claim detection (which parts are facts and which are opinions)
-> Detects factual claims and cross references it using google factcheck api to generate trust_score
-> Check_worthiness detection(2 pass BART-MNLI)
-> Source Credibility
Aggregator
-> the combined score of both modules give reliability score
-> reliability = f(bias, trust, sensationalism,framing)

Dual-Pipeline Architecture

Two Modules, Two Questions

IndiaGround evaluates news through a dual-pipeline system: one module examines how the story is written, and the other verifies what the story claims.

unBIAS Module

Analyzes HOW news is written

- Sentiment analysis
- Bias classification
- Framing detection

ClaimBuster Module

Analyzes WHAT is claimed

- Claim verification
- Evidence retrieval
- Fact-checking

Most current systems rely solely on sentiment analysis, but IndiaGround uses a dual-pipeline architecture that evaluates both HOW news is written and WHAT it claims.

Transparent Analysis at the Word Level

IndiaGround makes bias detection easy to audit by exposing the exact words and sources behind each judgment.

Flagged Tokens

IndiaGround flags individual biased words with a 90-term Indian media dictionary, rather than returning only a label.

Users can see exactly which words triggered each bias flag, making the analysis fully transparent.

Indian Credibility Layer

The system uses a 136-outlet credibility tier database to reflect the historical reliability of Indian publishers.

This adds outlet-level context when evaluating new content.

Pipeline Overview

Data Ingestion Pipeline

A clean, automated workflow that collects, filters, enriches, and stores news content for downstream analysis.

Built for high-volume news monitoring with automated de-duplication, entity extraction, and structured storage.

Source Collection

Collects articles every 30 minutes from Inshorts, Google News, and direct Indian outlet feeds.

Deduplication

Uses SHA-256 for exact matching and semantic analysis to remove 92% of near-duplicates.

Entity Recognition

Identifies people, places, and organizations with named entity recognition.

Storage

Stores processed content and metadata for downstream analysis.

Each stage is standardized to keep the pipeline reliable, scalable, and easy to audit.

The unBIAS Module: How News Is Framed

How the module evaluates sentiment, detects bias, and applies India-specific framing signals to make framing patterns more transparent.

Core analysis flow

1

Sentiment Analysis

Combines VADER (Valence Aware Dictionary and sEntiment Reasoner) for headlines with RoBERTa (Robustly Optimized BERT Pretraining Approach) for body-text analysis

2

Bias Classification

Uses BART-MNLI (Bidirectional and Auto-Regressive Transformers) for zero-shot detection of five bias types

3

India-Specific Framing

Detects labels such as "Pro-government," "Anti-government," "Communal," and "Nationalistic" using localized context

Five Bias Types Detected

- Political Bias
- Sensationalism
- Loaded Language
- Framing
- Omission

The ClaimBuster Module: Analyzing WHAT Is Claimed

Uses a three-stage pipeline to identify verifiable claims, retrieve supporting evidence, and produce per-claim verdicts.

Step 1

Check Worthiness

A local BART-MNLI model identifies which sentences contain verifiable factual claims versus opinions.

Step 2

Retrieve Evidence

DuckDuckGo searches multiple sources for supporting or refuting evidence.

Step 3

Verify Truth

Google Fact Check Tools API performs final truth verification against established fact-checking databases.

Output

Per-Claim Verdicts

Produces clear verdicts: "Supports", "Refutes", or "Not Enough Information".

RELIABILITY MODEL

Reliability Scoring Formula

The reliability score (R) combines four weighted components into a single percentage-based credibility metric.

$$R = [(1 - B) \times 0.35 + T \times 0.35 + (1 - S) \times 0.15 + (1 - F) \times 0.15] \times 100$$

Mathematical Foundation

In this formula, bias and trust each account for 35% of the score, so they have the strongest influence.

The remaining 30% is split between sensationalism and framing, which act as quality checks on how the claim is presented.

Key takeaway: the model rewards trustworthy, balanced claims while reducing the impact of bias, exaggeration, and distorted framing.

35%

Bias Inversion

This component uses 1–B: the lower the bias, the higher the contribution to reliability. A claim that is less biased earns a stronger score because it is more likely to present facts rather than persuasion.

35%

Trust Score

This is the direct confidence signal in the source or evidence quality. Higher trust raises the final score because well-supported claims should be rewarded most heavily.

15%

Anti-Sensationalism

This uses 1–S to reduce the score impact of exaggerated, emotional, or attention-grabbing wording. Less sensational language suggests the claim is more grounded and less manipulative.

15%

Framing Neutrality

This also uses 1–F to reward balanced framing. When a claim avoids one-sided presentation, it improves reliability because the information is less distorted by selective emphasis.

Technology Stack and Academic Foundation

Built for high-performance NLP processing and grounded in established research, this architecture combines modern infrastructure with a rigorous academic basis.

Technical Infrastructure

FastAPI Backend

Python 3.13 powers asynchronous, high-performance request handling.

AI Core

spaCy handles NLP tasks, supported by transformer models for deeper learning.

Database

PostgreSQL 18 with JSONB stores structured NLP outputs efficiently.

Infrastructure

Docker Compose, Celery, and Redis manage orchestration and background tasks.